# A Comparative Study of Generative Models: DCGAN Vs DDPM for Face Image Generation on the CelebA Dataset

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***Abstract*— Generative models have been successful in image synthesis tasks, and among them we can state Generative Adversarial Networks (GANs) and Diffusion Models. This paper explores practical trade-offs between these two different approaches, and this by comparing a Deep Convolutional GAN (DCGAN) against a Denoising Diffusion Probabilistic Model (DDPM). We used the CelebA dataset to train both models with the aim of generating human faces at 64x64 resolution. Then, we made our evaluation using the Fréchet Inception Distance (FID), the Inception Score, in addition to a manual evaluation, and results revealed a remarkable contrast: DCGAN is able to produce recognizable faces, but it struggles with training instability and lacks finer details. As for the DDPM, it proved to have much more reliable results via sharper and more realistic images. These findings highlight the reason behind adopting diffusion models as a new standard for high-fidelity image synthesis.**

**Keywords—Diffusion Models, Denoising Diffusion Probabilistic Models (DDPM), Deep Convolutional Generative Adversarial Networks (DCGAN), Generative Adversarial Networks, Image Synthesis, Face Image Generation, CelebA Dataset, Fréchet Inception Distance, Inception Score**

## I. Introduction

In computer vision and artificial intelligence, image generation has become an important task to shed some light on, especially as modern applications depend highly on realistic visual content. In this context, generative models were introduced to support data augmentation, creative media, identity simulation, and privacy-preserving analysis. These models need to not only generate images that resemble the ones they have been trained on, but also ensure diversity, sharp details, and natural vision in appearance.

Among the strongest approaches we have Generative Adversarial Networks (GANs) and Diffusion Models. The first approach (GANs) has dominated image synthesis for several years, but recent diffusion-based approaches have shown higher visual fidelity and better training stability.

In this study, we conduct a comparison between the classical Deep Convolutional GAN (DCGAN) and a Denoising Diffusion Probabilistic Model using the CelebA dataset. The objective is to assess empirically the difference between the two models in training behavior, visual realism, and quantitative image-quality metrics.

## II. Literature Review

Early comparative studies on generative models have been concentrated on understanding the main trade-offs between probabilistic latent models and adversarial learning. Chaturvedula and Battineedi (2024) focused on GANs against Variational Autoencoders (VAEs) using the MNIST dataset and found that GANs were better in producing sharper and more coherent images, while VAEs showed a good stability and were better at capturing the latent structure. Their results emphasize a recurring pattern which consists of the excellence of GANs in perceptual sharpness but their requirement of careful tuning to avoid blur and artifacts.

Shifting to a diffusion-based synthesis, Shen et al. (2024) compared diffusion models to adversarial networks using the MNIST and CIFAR-10 datasets. Results highlighted the smoothness of generated models and their ability to generate higher-quality samples at later iterations, meanwhile GANs were much faster but vulnerable to instability. This suggests that adversarial dynamics can result in a mode collapse or a divergence when dealing with unstable training conditions.

Beyond general benchmark datasets, diffusion models demonstrated a high performance even in domain-specific tasks. Indeed, Sequeira et al. (2024) conducted a comparison between DCGAN and a U-net diffusion model to generate synthetic tree-species images using a dataset of 2,178 wood cross-section images with 18 species. The study showed that the diffusion-based approach achieved lower FID and IS scores and improved classification performance for under-represented categories. Their findings suggest that diffusion is better than the adversarial generative strategy in preserving high-frequency texture information, which is aligned with improvements perceived in visual fidelity.

Another work has explored a hybrid mechanism that combines both paradigms. In fact, Liu et al. (2024) suggested a diffusion-driven GAN inversion for face image generation that leverages diffusion to reconstruct accuracy while maintaining the controllability and sampling efficiency of pretrained GAN models. This technique is reminiscent of the ongoing trend in generative research, where diffusion is highly adopted as a mechanism that aims to stabilize the adversarial results instead of replacing GANs entirely.

Taken all together, the literature suggests that there has been a point of transition reached in generative models. GANs currently retain their efficient aspect and ability to sample in a controllable manner, while recent diffusion models have demonstrated greater stability and detail. Therefore, these findings motivate a direct empirical comparison between a classical DCGAN and a DDPM to synthesize face images.

## III. Materials and Methods

### A. Dataset Description and Preprocessing

To conduct our experiment, we used the CelebA (CelebFaces Attributes) dataset. It contains a set of 202,599 images including 10,177 distinct identities. This ensures a rich variety of expressions, lighting, and backgrounds. Original images are 178x218 pixels and pre-centered around the face.

To ensure effective and consistent training of our models, we performed a processing pipeline on all the images, in which we center-cropped each image into a square and resized it to 64x64 pixels to align with the input requirements of both the DCGAN and the DDPM. Finally, we converted the resulting images into tensors and normalized them to the [-1, 1] range.




**Figure 1:** Example of image preprocessing applied in this work. The original CelebA image (left) and the corresponding preprocessed image resized to 64×64 (right).

### B. Generative Adversarial Networks (GANs)

Generative Adversarial Networks are called "adversarial" because their training is based on a minimax game between two neural networks: the generator G and the discriminator D. The generator takes noise vectors as input and converts them to candidate images. These generated images are given to the discriminator that has also been fed with the real images as another input, and it aims to distinguish between them and the ones produced by the generator. The discriminator is updated first to increase its ability to classify images, then the generator is updated via the discriminator's feedback to be able to output better images that will be harder to distinguish. Overall, the minimax game is described by the discriminator trying to maximize its accuracy to correctly differentiate between real and generated images, while the generator tries to minimize the discriminator's performance by fooling it with hardly distinguishable outputs.

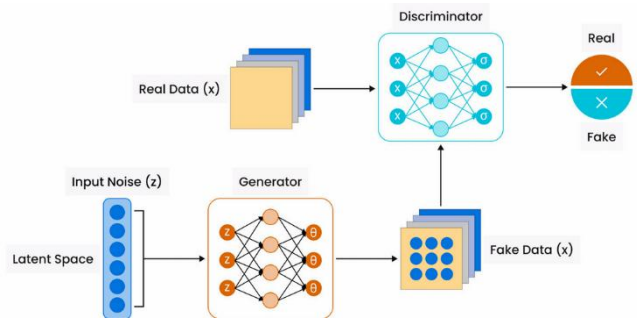



**Figure 2**: A Generative Adversarial Network's Architecture. The generator transforms latent vectors into candidate images, and the discriminator evaluates them against real samples to guide adversarial learning.

**Source**

#### 1) Deep Convolutional GAN (DCGAN)

DCGAN adapts the vanilla GAN into convolutional architectures by replacing fully connected layers with strided convolutions and transposed convolutions. This guarantees better spatial coherence and gives the model the ability to learn hierarchical visual features more effectively. DCGAN has become a common baseline to synthesize images, and it is particularly suitable to the CelebA dataset we chose to work on, where the model can reliably learn global facial structure.

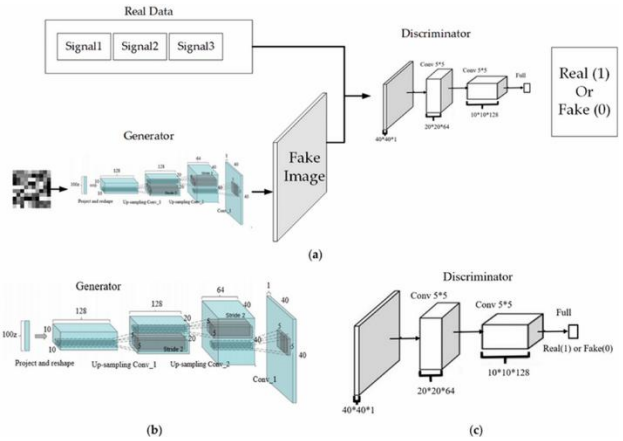



**Figure 3:** DCGAN architecture. (a) Overall adversarial setup in which the discriminator distinguishes real from generated samples. (b) Generator network composed of projection, reshaping, and transposed-convolution upsampling layers. (c) Discriminator network using successive convolutional layers to classify each image as real (1) or fake (0).

**Source**

#### 2) DCGAN Generator

The generator maps a latent vector $z \in \mathbb{R}^{100}$ to a $64 \times 64$ RGB image via a set of upsampling blocks. Each block represents a transposed convolution followed by Batch Normalization and ReLU activation. This progressively increases the spatial resolution from $4 \times 4$ to $64 \times 64$. A final Tanh layer is used to put pixel intensities in a $[-1,1]$range. This architecture allows smooth gradient flow and encourages the formation of sharp features at small resolutions.

### 3) *DCGAN Discriminator*

The discriminator performs inversely: it downsamples images by applying strided convolutions and extracts hierarchical features. It also uses LeakyReLU activations to reduce the risk of dead gradients and Batch Normalization for more stable intermediate feature distributions. Its final output is passed through a linear layer to estimate a single probability of authenticity. The discriminator is indeed effective, but it can quickly dominate training if it is not properly regulated, which requires careful balancing of learning dynamics.

## C. *Diffusion Models*

Diffusion models are a class of generative models that synthesize data by learning the reverse of a gradual noising process. During training, noise is progressively added to the original data through a forward diffusion process, while a neural network is trained to iteratively remove this noise in the reverse process to recover the underlying data distribution. Unlike adversarial models, diffusion models do not rely on a minimax game, which results in more stable training behavior and improved coverage of the data distribution. Recent studies have highlighted the effectiveness of diffusion models in generating high-quality images, particularly in comparison to GAN-based approaches, albeit with higher computational cost due to their iterative sampling procedure.

### 1) *Denoising Diffusion Probabilistic Model*

In this work, we adopt Denoising Diffusion Probabilistic Models (DDPM), which perform image generation through a forward process that progressively adds Gaussian noise to the data and a reverse process that iteratively removes this noise to recover a clean image.

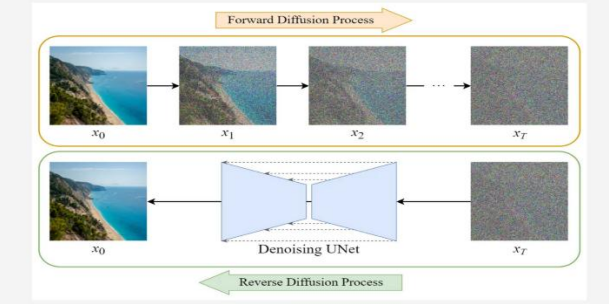



**Figure 4:** Overview of the diffusion process used in DDPM. The forward process progressively adds Gaussian noise to the original image, while the reverse process employs an ε-UNet to iteratively denoise the image and generate a clean sample.

**Source**

DDPM employs a UNet-based architecture to predict the noise added at each diffusion step, and the model is trained by minimizing the mean squared error (MSE) between the predicted noise and the true noise.

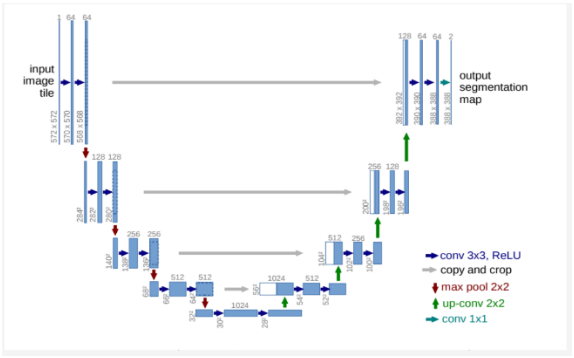



**Figure 5:** UNet-based architecture used as the ε-network in the DDPM model. The network follows an encoder–decoder structure with skip connections to predict the noise component at each diffusion timestep.

**Source**

We optimized the model using the Adam optimizer with a learning rate of 0.0002 over a duration of 80 epochs. Within the DDPM framework, we parameterized the reverse diffusion process using an $\varepsilon$-UNet (enhanced UNet), designed to predict the specific noise component added to an image at any given timestep. This network takes both the noisy image and a corresponding timestep embedding as inputs to generate an estimate of the noise. The primary training objective was to minimize the mean squared error (MSE) between this predicted noise and the actual noise sampled during the forward diffusion process. Following standard practice, we defined the diffusion process over T = 1000 timesteps, and to handle these computational demands, all experiments were carried out in a GPU-enabled environment.

## D. *Evaluation Metrics*

To quantitatively evaluate the quality and diversity of the generated images, we employed: Fréchet Inception Distance (FID) and Inception Score (IS).

### 1) *Fréchet Inception Distance (FID)*

Mathematically, FID gives the distance between feature distribution of real images and generated ones. It is defined as follows:

$$FID = \left|\left|\mu_r - \mu_g\right|\right|^2 + Tr\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{\frac{1}{2}}\right) \quad (1)$$

with $(\mu_r, \Sigma_r)$ and $(\mu_g, \Sigma_g)$ are respectively the means and covariances of Inception features for real and generated samples.

A low FID value suggests a higher visual quality, which means that the generated images occupy a similar distribution as the real ones in feature space.

The intuition behind the Fréchet Inception Distance is to evaluate how close synthetic faces are to real faces when it comes to sharpness, texture fidelity, distribution similarity, and sample diversity.

If the model produces blurry outputs or repetitive images, the covariance term increases to penalize the score.

*2) Inception Score (IS)*

Mathematically, the Inception Score assesses the clarity and diversity of generated images relying on the conditional label distribution $p(y \mid x)$ that the Inception classifier predicts. It is defined as shown below:

$$IS = exp\left(E_x\left[ KL\left(p(y|x) \parallel p(y)\right)\right]\right) \quad (2)$$

where p(y) is the marginal class distribution and KL is the Kullback–Leibler divergence.

Better quality is explained by a higher IS, because having strong confidence and diversity in the class usage increases the Inception Score.

Intuitively, clear and realistic images push the classifier to assign confident labels which improves the IS significantly. Additionally, if there is a variety in images, and they are not repetitive, then the marginal distribution spreads across classes which also enhances the IS. This score might significantly decrease if outputs are blurry or identical.

## IV. Results and Discussion

In this section, we present our experimental results after training both DCGAN and DDPM on the CelebA dataset. Our evaluation was based on quantitative and qualitative visual analysis to assess image quality, diversity, and realism. A discussion of results is provided to highlight strengths and weaknesses of each generative method.

### *A. Quantitative Evaluation*

For the evaluation of the quantitative performance of the proposed models in terms of quality and diversity of the resultant images, Fréchet Inception Distance (FID) and Inception Score (IS) metrics are used. Table 1 sheds light on the results for DCGAN and DDPM models.

TABLE 1. Quantitative Comparison of DCGAN and DDPM on CelebA

| Model | FID ↓ | IS ↑ |
|---|---|---|
| DCGAN | 73.63 | 1.85 ± 0.13 |
| DDPM | **56.79** | **1.90 ± 0.14** |

Lower FID values indicate that the generated images are closer to the real data distribution, while higher IS values reflect improved image quality and diversity. As shown in Table 1, DDPM outperforms DCGAN in both evaluation metrics, demonstrating its superior ability to generate realistic face images.

### *B. Qualitative Comparison of Generated Images*

The qualitative evolution of generated images across training epochs is illustrated in Figs. 4 and 5 for DCGAN and DDPM, respectively. At early training stages (Epoch 1), both models generate highly noisy and incoherent outputs, indicating that meaningful image structure has not yet been learned.

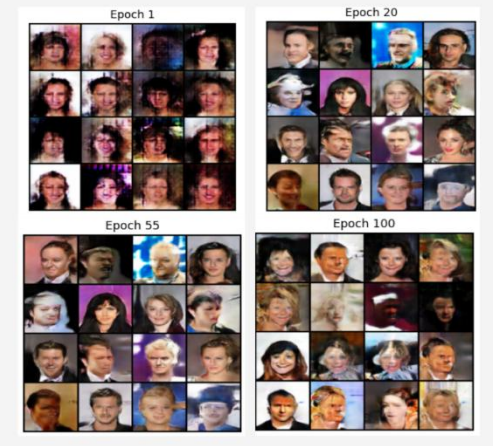



**Fig. 4.** Evolution of images generated by DCGAN at different training epochs (Epoch 1, 20, 55, and 100).

As training progresses, DCGAN begins to produce recognizable facial patterns; however, the generated images remain blurry and lack fine-grained details, even at later epochs. Artifacts and inconsistencies in facial features are still visible at Epoch 100, suggesting limitations in capturing high-frequency details and stable global structure.

This degradation in the performance of the DCGAN in later epochs of training is mostly driven by the instability of the adversarial training process. With the growing strength of the discriminator, the generator faces weaker and noisy gradient guides, which may counteract any potential improvements. There arises an imbalance, often resulting in a mode collapse, in which the generator fails to produce diverse images but often delivers similar images repeatedly. This practice may not really enhance image quality but rather reduce the realism of images produced.

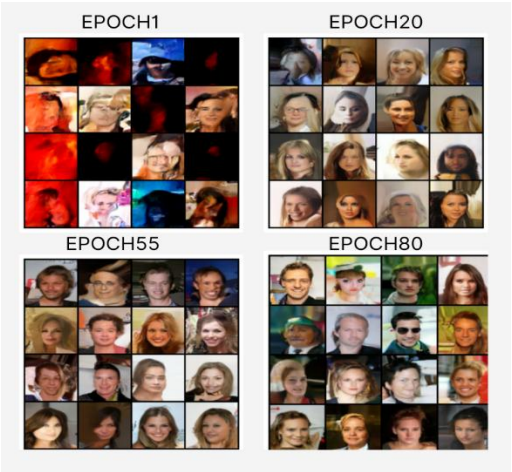



**Fig. 5.** Evolution of images generated by DDPM at different training epochs (Epoch 1, 20, 55, and 80).

Conversely, DDPM shows a more gradual consistent improvement in terms of the visual quality over epochs. By Epoch 55 and Epoch 80, the generated images are characterized by sharper facial features, improved texture consistency, and more coherent overall structure. This step-by-step approach

of removing noise progressively helps refine image details, which results in more realistic and visually appealing samples.

### C. Training Stability and Model Behavior

The training behavior of DCGAN was marked by a clear instability. Throughout the process, the fluctuating losses of both the generator and discriminator highlighted the constant struggle to maintain a perfect equilibrium between the two networks. As training continued, this delicate balance often shifted, which hindered the generator's ability to capture meaningful facial representations. This imbalance frequently manifested as blurred outputs or a drop in sample diversity, a phenomenon that became even more evident in later epochs. These results underscore how sensitive DCGAN is to both the length of training and the specific choice of hyperparameters.

On the other hand, the DDPM model proved to be remarkably stable and consistent throughout its training. Because optimization is built on minimizing the mean squared error between the predicted and actual noise, the loss curves showed a smooth, steady decline rather than the erratic jumps we saw with the DCGAN. As a result, the DDPM demonstrated a very dependable convergence, with the quality of the generated faces improving predictably as the epochs went by.

## CONCLUSION

This paper presented a comparison between DCGAN, a GAN-based model, and DDPM, a diffusion-based model, for the synthesis of faces on the CelebA dataset. By comparing the experiments on both models, it was seen that DDPM performs better than DCGAN with respect to the quality, realism, and training stability of the generated faces. Even though DCGAN models are able to generate faces that can be identified as faces, their adversarial training process reduces their stability and efficiency in the long run, whereas DDPM models can produce faces that are clearer and are obtained through a stable process.

As a result of limited availability of powerful GPUs, training had to happen with limited computational resources, thus restricting the number of epochs and model complexity. Yet, the diffusion model outperformed the generative adversarial model, which indicates robustness and power of the process. In the future, improving efficiency and using more robust computational power would show benefit in improving creation quality and resolution.

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